Are Action Units Priors or Pathways? A Diagnostic Study of Temporal Valence-Arousal Estimation

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Abstract

Facial Action Units and continuous valence-arousal are complementary but distinct representations of facial affect. While temporal valence-arousal estimation is well studied, the role of Action Units within this task remains unclear: should they be omitted, used as auxiliary training priors, supplied as predicted features, or used as the dominant pathway for valence-arousal prediction? This question is addressed through a systematic architectural study on the overlapping valence-arousal and Action Unit subset of the Aff-Wild2 dataset. The five AU-routing configurations share the same ResNet-50 backbone, Transformer temporal encoder, and training protocol; separate controls vary geometry and temporal-window length. In the single-run comparison, prediction without Action Unit supervision reaches a mean concordance correlation coefficient of 0.3339. Action Unit feature fusion improves this to 0.3472 with concatenation and 0.3643 with gating, while Action Unit-mediated prediction reaches 0.3688. The strongest configuration, auxiliary Action Unit supervision with direct temporal prediction and no MediaPipe geometry, reaches 0.3887 for a single run and 0.3975 ± 0.0133 across five seeds. Frame-level pathway diagnostics extend the analysis beyond the overall concordance correlation coefficient evaluation by comparing routing behaviour across distance from the neutral valence-arousal point, temporal stability, and facial Action Unit activation regimes. These findings indicate that Action Units are most effective as representation-shaping priors rather than as universal valence-arousal pathways.

Keywords: Action Units, Valence-Arousal, Temporal Affect Recognition, Multi-Task Learning, Auxiliary Supervision

1 Introduction

The study of human emotion involves multiple complementary representations. Action Units (AUs), anatomically defined facial muscle movements, provide a low-level, interpretable description of expression mechanics. In contrast, dimensional models using valence (positive-negative) and arousal (low-high activation) capture global affective state. Temporal VA prediction from facial video has been extensively studied, with recent systems employing recurrent, convolutional, transformer-based, and audio-visual temporal encoders. Thus, the central question is not whether temporal modeling benefits VA prediction, but how interpretable AU structure should be incorporated into temporal VA systems.

Existing approaches typically predict AUs and VA independently or use AUs as auxiliary supervision, but seldom examine whether VA prediction should be routed through estimated AUs. This design is intuitive because AUs describe facial muscle activations. However, accurate VA estimation may also depend on temporal context, facial appearance, expression intensity, head motion, and subtle cues outside the limited supervised AU set. Requiring predicted AUs to form a mandatory intermediate representation may therefore improve interpretability while discarding information needed for accurate VA estimation. This study frames AU integration as a routing problem: whether AUs shape training, enter the VA regressor as predicted features, or become the dominant intermediate representation. This is an architectural prediction question, not a causal analysis of whether facial actions determine affect, which would require interventions and causal assumptions unavailable from observational annotations. On the Aff-Wild2 VA+AU overlap split, controlled architectural comparisons cover VA-only prediction, direct VA prediction with auxiliary AU supervision, AU feature fusion, and AU-mediated prediction. Compact MediaPipe [Lugaresi et al., 2019] geometry and a shorter temporal window serve as controls, while diagnostics compare aggregate and frame-level errors across affect and facial-action regimes.

The key contributions of the paper are:

- AU supervision improves temporal VA estimation without requiring predicted AUs in the inference route: every AU-aware variant exceeds VA-only prediction, while AU-aux direct is strongest in the controlled architectural comparison [1].
- The supervised 12-AU probability vector is informative but incomplete as a universal VA route. Concatenation, gating, and mediation improve on VA-only prediction yet remain below AU-aux direct, indicating that the shared temporal representation retains additional VA-relevant information.
- The value of AU routing is condition-dependent. Five-seed evaluation and frame-level diagnostics show that AU-aux direct is strongest near neutral and during stable affect, whereas mediation retains selective value for upper-face activation regimes.

2 Related Work

Continuous affect recognition is commonly formulated through the widely used two-dimensional valence-arousal model [Russell, 1980], whereas facial expression analysis often relies on the Facial Action Coding System (FACS) [Ekman and Friesen, 1978]. Large-scale in-the-wild benchmark datasets such as Aff-Wild2 [Kollias and Zafeiriou, 2018] and ABAW [Kollias, 2022b] support evaluation across VA estimation, AU detection, and expression recognition under naturalistic conditions. These datasets introduce substantial variation in pose, illumination, occlusion, identity, and recording environment, making them important testbeds for modern affect recognition models.

Temporal modeling plays a central role in VA estimation from facial video. Recurrent networks, temporal smoothing, causal sequence models, transformers, and audio-visual fusion have all been used to improve over frame-level baselines [Meng et al., 2022, Zhang et al., 2021]. In these approaches, performance gains are primarily attributed to improved temporal representation learning, and the value of temporal context for VA prediction is well established. In contrast, the objective here is to hold the temporal backbone fixed and study the AU-to-VA routing mechanism as the primary experimental variable.

Multi-task learning across AUs, expressions, and VA is motivated by the intuition that lower-level facial signals can regularize higher-level affect prediction [Kollias, 2022a, Das et al., 2024]. ABAW systems and related multi-task approaches [Kollias, 2022b, Kollias, 2022a] often benefit from joint AU and VA supervision, making AU co-supervision a common design component. However, prior studies rarely disentangle three qualitatively different uses of AUs: auxiliary supervision that shapes encoder representations during training, predicted AU features that augment the VA head at inference, and enforced intermediate pathways that require VA to be inferred through AU activations. This distinction has direct architectural implications, because each strategy makes a different assumption about whether the supervised AU space is sufficiently expressive to support continuous VA prediction.

Geometric and landmark features offer a complementary interpretable prior. Compact facial and behavioral descriptors, including landmarks, pose, gaze, and AU features, have been used in related affective-state recognition settings [Das and Dev, 2025]; tools such as MediaPipe [Lugaresi et al., 2019] further enable lightweight

[1] Code is publicly available at `https://github.com/rijju-das/AU-Routing-VA` to support reproducibility and further research.

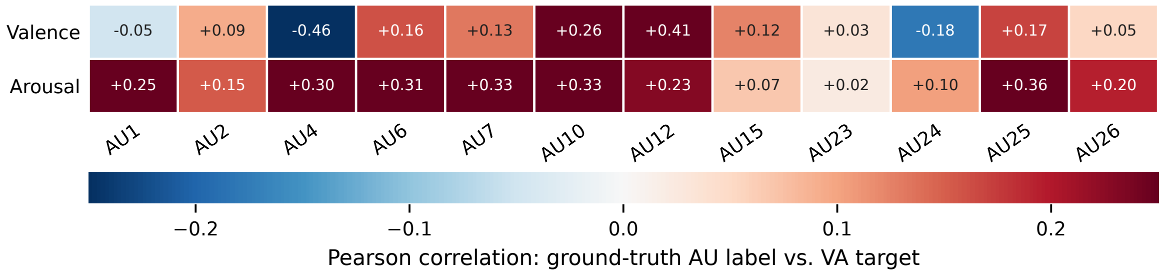

Figure 1: Frame-level Pearson correlations between each binary ground-truth AU label (columns) and the continuous valence and arousal annotations (rows) on the Aff-Wild2 overlap training set. Each cell is an AU-to-VA correlation computed across paired annotated frames.

facial geometry extraction. However, their value as fused priors for strong RGB-based temporal VA models remains unclear. Such features may provide explicit facial structure, but they may also be redundant with learned appearance representations or introduce noise when landmark detection is unreliable. MediaPipe geometry is therefore included as a controlled ablation rather than assumed to improve VA prediction.

3 Diagnostic Framework

3.1 Data and Setup

All experiments are conducted on the Aff-Wild2 VA+AU overlap subset, where each training sample contains both continuous valence-arousal (VA) annotations and binary labels for 12 facial Action Units: AU1, AU2, AU4, AU6, AU7, AU10, AU12, AU15, AU23, AU24, AU25, and AU26. The split contains 295 training videos and 65 validation videos. Unless otherwise stated, videos are sampled as 16-frame windows with a stride of 8. Frames are resized to 112×112 and normalized using ImageNet statistics. A compact temporal-context ablation uses 8-frame windows while keeping the remaining preprocessing and model configuration unchanged. Figure 1 reports the frame-level Pearson correlation between each binary ground-truth AU label and each continuous VA target (AU-to-VA correlations) on the overlapping training set. Their heterogeneous signs and absolute values, ranging from 0.02 to 0.46, show that individual AUs carry related but incomplete linear information about VA. The figure therefore motivates an empirical test of AU routing; it does not establish whether the combined AU vector is sufficient for VA estimation.

3.2 AU Routing Variants

All model variants use the same shared visual-temporal encoder. Given a window of T RGB frames, a ResNet-50 backbone pretrained on ImageNet extracts per-frame visual features. The resulting feature sequence is passed to a two-layer vanilla self-attention Transformer encoder [Vaswani et al., 2017], producing frame-wise temporal representations $\mathbf{f}_t \in \mathbb{R}^{256}$. Across the routing comparisons, the encoder architecture and representation dimensionality remain fixed, while the use of AU information varies. The AU head maps this representation to 12 frame-level activation probabilities, $\hat{\mathbf{u}}_t = \sigma(h_{\mathrm{AU}}(\mathbf{f}_t)) \in [0,1]^{12}$, while the direct VA head predicts a two-dimensional valence-arousal coordinate, $\hat{\mathbf{y}}_t^{\mathrm{direct}} = \tanh(h_{\mathrm{VA}}(\mathbf{f}_t)) \in [-1,1]^2$. The $h.$ terms denote independently learned two-layer feed-forward heads. Figure 2 summarizes how the shared temporal representation and predicted AUs are used across the five comparison variants.

The variants span a continuum from no AU use to AU-dominant routing. **Pure VA-only** ignores AU labels and predicts VA directly from $\mathbf{f}_t$. **AU-aux direct** is a multi-task configuration where the AU and VA losses jointly shape the shared encoder during training, while VA is predicted directly from $\mathbf{f}_t$ rather than from $\hat{\mathbf{u}}_t$. Unlike AU-aux direct, the remaining three AU-aware variants explicitly introduce predicted AUs into VA estimation. In **AU concatenation**, the VA head receives both the temporal representation and predicted AU

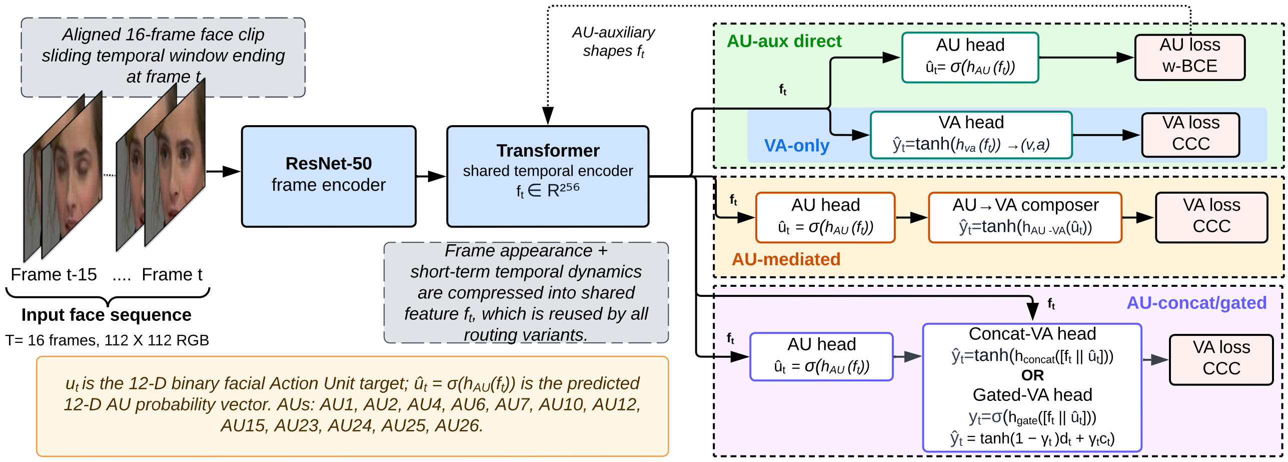

Figure 2: Five comparison variants sharing the same ResNet-50 + Transformer backbone ($\mathbf{f}_t \in \mathbb{R}^{256}$, $T = 16$ frames). The blue branch represents the pure VA-only baseline; adding the auxiliary AU head yields AU-aux direct. AU-mediated VA estimates VA primarily from predicted AUs, while AU concatenation and AU-gated fusion introduce them explicitly at the VA head. Repeated loss connections and the scalar residual path are omitted for visual clarity.

probabilities:

$$\hat{\mathbf{y}}_t^{\text{concat}} = \tanh\left(h_{\text{concat}}\left([\mathbf{f}_t; \hat{\mathbf{u}}_t]\right)\right). \tag{1}$$

In **AU-gated fusion**, a two-dimensional direct VA score $\mathbf{d}_t \in \mathbb{R}^2$ derived from $\mathbf{f}_t$ and an AU-derived VA score $\mathbf{c}_t \in \mathbb{R}^2$ derived from $\hat{\mathbf{u}}_t$ are combined using a frame-level, per-dimension gate $\gamma_t \in [0,1]^2$:

$$\gamma_t = \sigma\left(h_{\text{gate}}\left([\mathbf{f}_t; \hat{\mathbf{u}}_t]\right)\right), \quad \hat{\mathbf{y}}_t^{\text{gated}} = \tanh\left((1-\gamma_t)\mathbf{d}_t + \gamma_t \mathbf{c}_t\right). \tag{2}$$

In **AU-mediated VA**, the prediction passes primarily through the predicted AU probabilities:

$$\hat{\mathbf{y}}_t^{\text{AU}} = \tanh(h_{\text{AU2VA}}(\hat{\mathbf{u}}_t) + \alpha r_t), \tag{3}$$

where $r_t = \frac{1}{256}\sum_{j=1}^{256} f_{t,j}$ is a scalar temporal-feature summary broadcast to both VA dimensions, and α is a learnable scalar initialized to 0.01. This produces an AU-structured route with a low-dimensional residual rather than a strictly AU-only pathway.

4 Experiments and Results

4.1 Training and Evaluation Setup

Training uses AdamW for 20 epochs (learning rate 5×10^{-5}; weight decay 10^{-5}), cosine scheduling, unit-norm gradient clipping, and batches of 8 accumulated over four steps (effective batch 32). ResNet-50 is frozen for 10 epochs and then unfrozen with the learning rate doubled. For $d \in \{v, a\}$, the concordance correlation coefficient (CCC) is

$$\rho_c^{(d)} = \frac{2\,\text{cov}(\mathbf{y}^{(d)}, \hat{\mathbf{y}}^{(d)})}{\text{var}(\mathbf{y}^{(d)}) + \text{var}(\hat{\mathbf{y}}^{(d)}) + (\mu_{y^{(d)}} - \mu_{\hat{y}^{(d)}})^2 + \epsilon}, \tag{4}$$

where $\mathbf{y}^{(d)}$ and $\hat{\mathbf{y}}^{(d)}$ are valid batch labels and predictions; cov, var, and μ are population moments; and $\epsilon = 10^{-8}$ provides numerical stability. The VA loss is $\mathcal{L}_{\text{VA}} = (1-\rho_c^{(v)}) + 1.5(1-\rho_c^{(a)})$, mildly upweighting arousal. AU detection uses weighted binary cross-entropy to address class imbalance, and AU-aware models optimize the sum of the AU and VA losses with unit task weights. The pure VA-only ablation sets the AU loss weight to zero. The best epoch for each run is selected using validation mean CCC, computed as the average of valence CCC and arousal CCC. AU quality is reported with macro F1 at threshold 0.5.

Geometry is a secondary control, not part of the AU-routing mechanism. This geometry is extracted using MediaPipe to test whether explicit facial shape and pose cues complement the learned RGB-temporal representation or alter the relative behaviour of the routing variants. Only "+ MediaPipe" variants receive a precomputed 40-D vector: 32 normalized coordinates from 16 landmarks, three pose proxies, four bounding-box values, and one validity flag. A learned gate adds its projected embedding residually to the RGB-temporal representation, and disk caching fixes these inputs across runs.

4.2 AU Role, Robustness, and Geometry Ablations

Table 1 compares the tested AU-integration strategies and the two control ablations for geometry and temporal context, forming a clear performance ladder. Pure VA-only prediction is weakest, reaching 0.3339 mean CCC. Adding predicted AUs as explicit VA features improves performance to 0.3472 with concatenation and 0.3643 with gating. AU-mediated VA reaches 0.3688. The best single run is AU-aux direct, reaching 0.3887.

Group	Variant	Mean CCC	CCC_v	CCC_a	AU F1
AU role	Pure VA-only, no AU loss	0.3339	0.2066	0.4613	-
	AU concatenation, no MediaPipe	0.3472	0.2269	0.4675	0.4440
	AU-gated fusion, no MediaPipe	0.3643	0.2851	0.4435	0.4478
	AU-mediated VA, no MediaPipe	0.3688	**0.2870**	0.4506	0.4334
	AU-aux direct, no MediaPipe	**0.3887**	0.2800	**0.4973**	0.4393
Geometry	AU-aux direct + MediaPipe	0.3574	0.2717	0.4430	0.4428
	AU-mediated VA + MediaPipe	0.3470	0.2462	0.4477	0.4323
Temporal	AU-aux direct + MediaPipe, $T = 8$	0.3456	0.2920	0.3993	**0.4529**

Table 1: Main validation results on the Aff-Wild2 VA+AU overlap split, grouped by the tested factor. All variants use 16-frame windows unless explicitly marked otherwise. AU-aware variants improve over pure VA-only prediction, and the strongest result is AU-aux direct.

Together, Table 1 and Figure 3 support the central claim. Every AU-aware variant improves over pure VA-only prediction, but the strongest use of AUs is not hard mediation. AU concatenation, AU-gated fusion, and AU-mediated VA progressively improve on VA-only prediction but remain below AU-aux direct, which exceeds AU-mediated VA without MediaPipe by +0.0199 CCC. Thus, AU supervision improves temporal representations, which retain VA-relevant information beyond predicted AU probabilities.

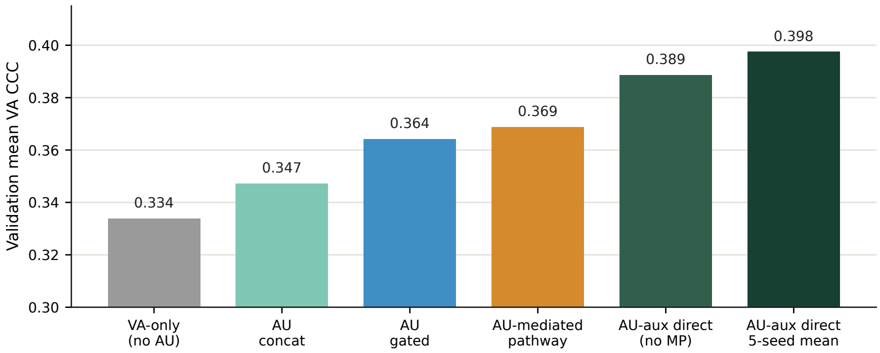

Figure 3: AU-role ablation ladder. The best single run is AU-aux direct at 0.3887 mean CCC; explicit AU routes improve over VA-only prediction but remain weaker. MP denotes MediaPipe.

Because single runs can overstate differences, Table 2 reports five-seed robustness for AU-aux direct and the matching MediaPipe variant. AU-aux direct, no MediaPipe obtains 0.3975 ± 0.0133 mean CCC, while AU-aux direct + MediaPipe obtains 0.3764 ± 0.0131. Across the same five matched seeds summarized in Table 2, the no-MediaPipe model wins in 4 out of 5 paired comparisons with a mean paired gain of +0.0211 CCC. Thus Table 2 shows that the main conclusion is not driven by a single favourable run.

The geometry and temporal rows in Table 1 show a negative but informative result. Compact MediaPipe geometry reduces VA performance for both routing conditions: AU-aux direct drops from 0.3887 to 0.3574

Variant	Mean CCC	CCC_v	CCC_a	AU F1
AU-aux direct, no MediaPipe, 5 seeds	**0.3975 ± 0.0133**	**0.3273 ± 0.0338**	**0.4678 ± 0.0180**	**0.4458 ± 0.0063**
AU-aux direct + MediaPipe, 5 seeds	0.3764 ± 0.0131	0.2925 ± 0.0166	0.4603 ± 0.0302	0.4376 ± 0.0059

Table 2: Five-seed validation robustness. AU-aux direct is stronger on average; over matched seeds, the no-MediaPipe model wins 4/5 paired comparisons with a mean paired gain of +0.0211 CCC.

mean CCC, and AU-mediated VA drops from 0.3688 to 0.3470. The temporal-window row further shows that the 8-frame MediaPipe variant is weaker than the corresponding 16-frame MediaPipe model for VA (0.3456 vs. 0.3574), although it yields the highest AU F1 (0.4529). This suggests that VA benefits from longer temporal context, while AU detection may be more local and frame-proximal. More importantly, the geometry experiment cautions against assuming that interpretable priors automatically improve a strong RGB-temporal encoder.

4.3 Pathway Diagnostics

Aggregate CCC identifies the winner, but not the mechanism. To test whether each AU route helps in different affect regimes, validation outputs are aligned by temporal-window position and their frame-level errors are compared between pathways. For models A and B, listed before and after "vs." in Table 3, Base CCC and Comp. CCC report their respective aggregate scores. ΔMAE is $\text{MAE}_B - \text{MAE}_A$, so positive values favour baseline A; win rate is the fraction of aligned predictions for which A has lower VA error. Ground-truth distance from neutral is $\delta_t = \sqrt{v_t^2 + a_t^2}$ and is grouped into near-neutral ($\delta_t < 0.15$), low ($0.15 \le \delta_t < 0.35$), medium ($0.35 \le \delta_t < 0.60$), and high ($\delta_t \ge 0.60$) bins. Table 3 summarizes the comparisons.

Comparison	Base CCC	Comp. CCC	ΔMAE	Win Rate
AU-aux direct vs. AU-mediated VA, no MP	0.3871	0.3675	+0.0012	0.503
AU-aux direct vs. AU-mediated VA, MP	0.3561	0.3461	+0.0327	0.624
AU-aux direct, no MP vs. MP	0.3871	0.3561	+0.0164	0.542
AU-mediated VA, no MP vs. MP	0.3675	0.3461	+0.0480	0.638
AU-aux direct, MP, $T = 16$ vs. $T = 8$	0.3365	0.3223	+0.0040	0.515

Table 3: Diagnostic comparisons; MP denotes MediaPipe and T the temporal-window length in frames. Positive ΔMAE favours the baseline. The first four use 545,327 aligned valid VA predictions from overlapping windows; the final comparison aligns VA predictions at the shared starting frame of each matched window.

To show whether the aggregate difference varies across conditions, the first comparison in Table 3 is further decomposed across affect and AU regimes in Figure 4. Although the overall difference between AU-aux direct and AU-mediated VA without MediaPipe is small (+0.0012), substantially larger differences emerge within individual subgroups. In this decomposition, AU-aux direct is especially favourable for negative/low-arousal frames (ΔMAE = +0.0342), near-neutral frames (+0.0244), frames at low VA distance (+0.0141), and temporally stable windows (+0.0122). Here, temporal stability denotes a mean ground-truth VA standard deviation below 0.05 within a window. These results suggest that the supervised AU activations may not fully characterize VA under these conditions. In contrast, the AU-mediated model remains favourable for some upper-face patterns, including upper-face-only frames, where at least one upper-face AU is active and lower-face AUs are inactive (−0.0333), and eye/cheek-only frames (−0.0153). AU mediation therefore remains informative in selected facial-action regimes, although it is not the strongest overall route for VA estimation.

5 Discussion

The results separate two roles for AUs that are often conflated in multi-task affect models: supervision and routing. AU supervision is useful because it encourages the shared temporal representation to encode facial-action structure. However, using the predicted AU vector $\hat{\mathbf{u}}_t$ as the primary information route for VA is more restrictive. The AU-aux direct model keeps VA prediction conditioned on the full temporal representation while

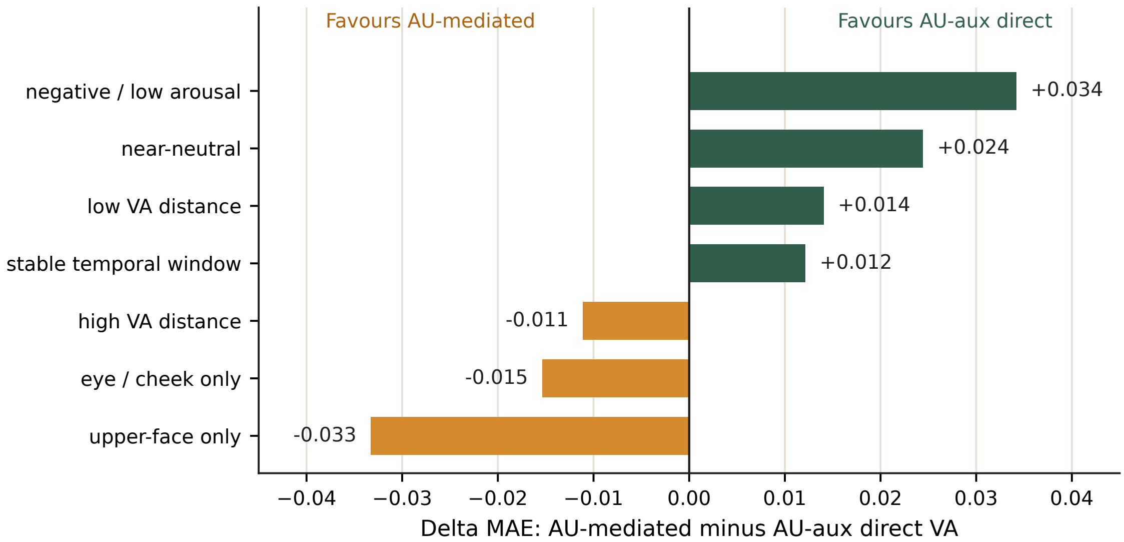

Figure 4: Grouped pathway diagnostics for AU-aux direct versus AU-mediated VA without MediaPipe. Positive ΔMAE favours AU-aux direct; negative values favour AU-mediated routing.

still using AU labels to shape that representation during training. This design preserves access to cues that fall outside the 12 supervised AUs, including expression intensity, facial appearance, person-specific dynamics, and sequence-level context. The concatenation and gated routing variants occupy an intermediate position between priors and pathways. They demonstrate that predicted AU probabilities can serve as useful VA features, particularly when a learned gate modulates the contribution of the AU-derived estimate. That neither variant surpasses AU-aux direct suggests that, in the tested setting, the expressiveness of the AU probability vector may be a stronger limitation than the choice of fusion mechanism. Once AUs have shaped the encoder, providing them explicitly at the VA head offers diminishing returns. These findings concern information flow within the trained models; they do not establish that AUs cause, mediate, or determine human affect, and the observed differences across affect and AU subgroups should be interpreted as associations rather than causal effects.

For the geometry control, the result applies only to the tested compact descriptor, not to facial geometry generally. The descriptor reduces VA performance in both routing conditions, and the no-MediaPipe model leads in four of five paired runs (Table 2). Static coordinates may be redundant with learned RGB shape cues or noisy under imperfect landmark fits. Richer dynamic geometry, including landmark velocities, distances, and symmetry, was not evaluated in this work. Therefore, the supported conclusion is limited to this descriptor and fusion setting.

The study is also limited to the Aff-Wild2 VA+AU overlap subset. Paired frame annotations enable controlled comparison, but the hierarchy may depend on the dataset's annotation protocol, subject and recording distribution, label reliability, and 12-AU inventory. AU type and count are fixed, so the experiments cannot attribute the benefit to particular units or determine whether a larger inventory would change the ranking. The evidence is therefore within-dataset rather than universal. Replication, cross-dataset transfer, and AU-subset ablations are needed to test generalization. The diagnostic asymmetry motivates hybrid architectures that apply AU-mediated routing selectively for high-confidence upper-face activations, while retaining direct prediction when AU evidence is weak or ambiguous.

6 Conclusion

This work tests AUs as auxiliary supervision, predicted features, and AU-dominant pathways for temporal VA estimation. Under controlled comparisons using the same ResNet-50 and Transformer on the Aff-Wild2 VA+AU overlap split, every AU-aware variant exceeds pure VA-only prediction (0.3339 mean CCC). AU-aux direct is strongest at 0.3887 for a single run and 0.3975 ± 0.0133 across five seeds, ahead of concatenation

(0.3472), gated fusion (0.3643), and mediation (0.3688). Diagnostics show its largest advantages near neutral and within temporally stable windows, while mediation retains selective value for upper-face activations. Within the evaluated dataset and 12-AU inventory, AUs are therefore valuable but conditional: they provide interpretable facial-action evidence yet do not form a complete route for contextual or weakly expressed affect. The findings support AUs as representation-shaping priors rather than universal VA pathways and motivate selective routing when AU evidence is reliable.

Acknowledgments

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